

Technical Datasheet - archN_robust (VBF-T5R1SINGLEMODEL-DSH-001)

> Case: t5_r1_singlemodel. Values mechanically taken from parameter set / simulation results and source-annotated; simulated scope stated honestly; cycle life not fabricated.

Field	Value	Source
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Rated capacity	5.0 Ah nominal (parameter set); 5.0374 Ah simulation-verified (1C DFN discharge)	`OKane2022:Nominal cell capacity [A.h]` / `cell/r5_final_archN_1c_dfn.json`
Nominal voltage / voltage window	3.71 V midpoint (1C discharge) / 2.5 - 4.2 V	`cell/r5_final_archN_energy_dfn.json:midpoint_voltage_v` / `OKane2022` cut-offs
Rated energy	18.37 Wh (1C discharge V-I integration)	`cell/r5_final_archN_energy_dfn.json:energy_wh`
Gravimetric energy density	533.18 Wh/kg (contract caliber: discharge energy ÷ layer-mass total, electrolyte excluded)	`cell/r5_final_archN_energy_dfn.json:energy_density`
Volumetric energy density	968.10 Wh/L (layer-stack volume caliber)	`cell/r5_final_archN_energy_dfn.json:energy_density_wh_l`
Maximum continuous discharge rate	1C (simulated; 5C capability not tested in this case — library protocol available)	`1C_discharge` protocol run
Fast-charge capability	4C charge at 45 °C ambient: max cell temperature 330.50 K (57.35 °C) ≤ 60 °C; anode potential min +0.0215 V → no lithium plating. CC-charge capacity not simulated	
Operating temperature range	Simulated conditions only: 25 °C ambient (1C discharge), 45 °C ambient (4C charge). Wider-range operability not simulated — stated honestly	
Cycle life	**Not simulated (would require aging-model runs)** — not provided rather than fabricated library `aging_1C_100cyc` protocol exists but not run in this case	
Safety determination	4C fast charge without plating: ✓; max temperature 57.35 °C ≤ 60 °C: ✓ (2.65 K margin)	`cell/r4_archN_4c_dfn.json`
DC internal resistance	15.89 mΩ (1C-discharge-derived: (V ₀ −V _{10%})/I _{1C})	`cell/r5_final_archN_energy_dfn.json:dcr_ohm`
Dimensions	65 mm × 1580 mm × 0.1848 mm (layer stack; shell thickness not provided)	`OKane2022:Electrode height/width [m]` + calc-energy thickness
Mass	34.46 g (electrode stack + CC + separator, contract caliber); 41.08 g including electrolyte at literature density 1.2 g·cm ⁻³	`cell/r5_final_archN_energy_dfn.json:mass_k`
Electrolyte / additives	LiPF ₆ EC/EMC-class base electrolyte, σ 3.5 S·m ⁻¹ , D 1.2×10 ⁻⁹ m ² ·s ⁻¹ , t ⁺ 0.55; recommended additives VC / FEC / DTD (qualitative, mace-screened)	`cell/r5_final_archN_energy_dfn.json:electrolyte`